

Access Right

7	6	5	4	3	2	1	0
P	DPL	S	E	ED/c	R/W	A	

→ A=0, Segment not accessed
A=1, Segment has been accessed

→ R=0, Code segment may not be read
R=1, Code segment may be read
W=0, data segment may not be written
W=1, data may be written

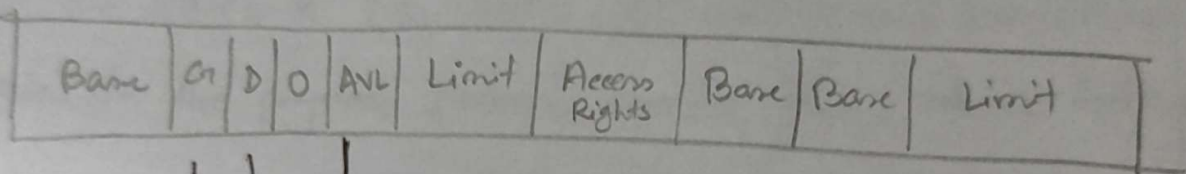
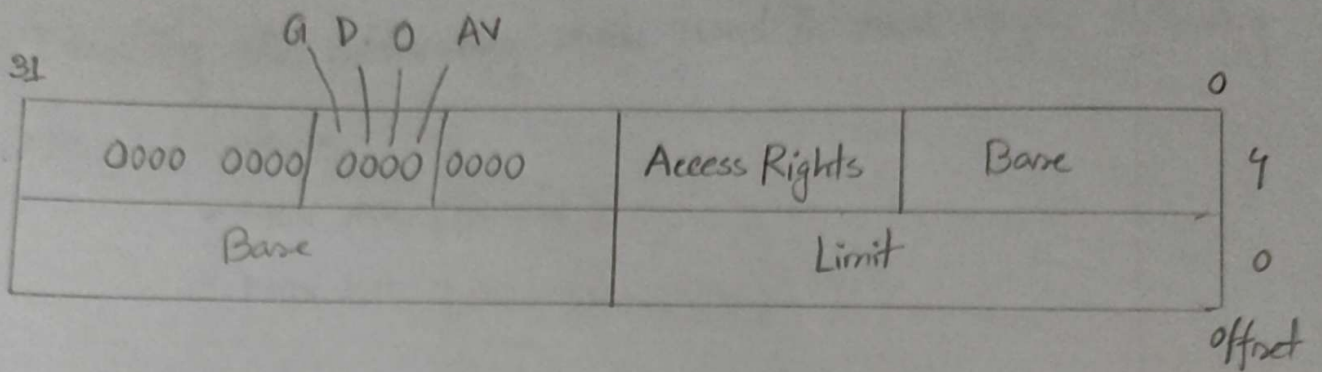
→ ED=0, Segment expands upward (data segment)
ED=1, Segment expands downward (stack segment)
C=0, ignore descriptor privilege level
C=1, Abide by privilege level

→ E=0, Descriptor describes a data segment
E=1, Descriptor " code segment

→ S=0, system descriptor
S=1, code or data segment

→ Sets the descriptor privilege level

→ P=0, Descriptor is undefined
P=1, Segment contains a valid base and limit



→ 0; segment AVL
 1; " not AVL
 D: 0; 16 bit
 1; 32 bit
 → G=0; limit = 10000H
 = 1; limit = 10000

Sec-3

A

1. Identify the addressing mode used in each of the following 8086 instructions

• MOV AX, [2690H]

• $\text{MOV AX, [BX+DI+200H]}$

If the values of $\text{DS} = 3000\text{H}$, $\text{DI} = 0090\text{H}$ and $\text{BX} = 0100\text{H}$ then calculate the physical address accessed by the above instructions.

From the given instructions,

MOV AX, [2690H] - Direct Addressing Mode (MOV reg, r/m)

$\text{MOV AX, [BX+DI+200H]}$ - Base plus index Addressing Mode.

($\text{MOV reg, [base+index+Displacement]}$)

Given that,

$\text{DS} = 3000\text{H}$

$\text{DI} = 0090\text{H}$

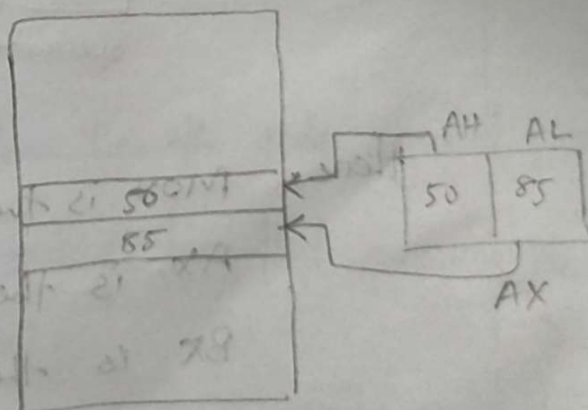
$\text{BX} = 0100\text{H}$

i) MOV AX, [2690H]

we get, $\text{DS} = 3000\text{H}$

offset = 2690H

$$\begin{aligned}\therefore \text{Physical address} &= (\text{Segment} \times 10\text{H}) + \text{offset} \\ &= 3000 \times 10\text{H} + 2690 \\ &= 32690\text{H}\end{aligned}$$



ii) `MOV AX, [BX+DI+200H]`

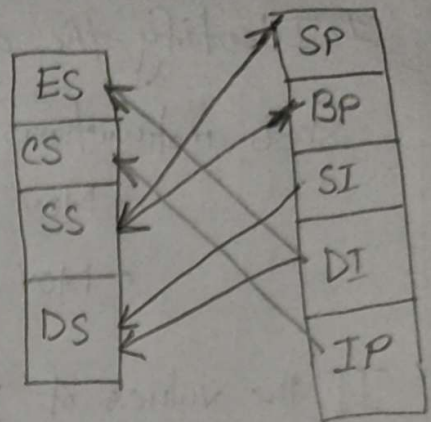
Physical address

$$= (\text{Segment} \times 10H) + \text{offset}$$

$$= (3000 \times 10) + (0100 + 0090 + 200)$$

$$= 30000 + 00390$$

$$= 30390H$$



2. Describe Register Relative addressing mode.

Register Relative addressing mode is shown as

`MOV reg, [base + displacement]`

`MOV AX, [BX + 200H]`

operation | destination | Source

Here, MOV is the operation

AX is the destination register

BX is the base register

200H is the displacement

Suppose, $DS = 3000H$

$BX = 0100H$

for physical address,

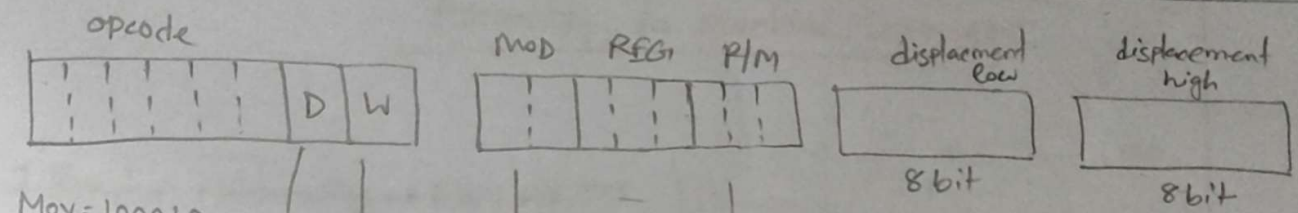
$$\begin{aligned} & (\text{Segment} * 10H) + \text{offset} \\ &= (3000 * 10) + (0100 + 200) \\ &= 30000 + 0300 \\ &= 30300H \end{aligned}$$

3. Consider the following scenario

- You must define priority of the descriptor privilege level (DPL) into the access right portion so that segment access will not be granted.
- Set a bit of access right portion that ensures CS (Code Segment) has valid address & limit.
- Also ensure the descriptor describes the code segment portion.
- Define a bit in the access right portion that confirms the data segment will not be read..

Your task is to compute 8 bits of Access Right Byte for 80386. Also analyze the details of each bit so that the above operations will perform correctly.

Chp-4



Mov = 100010
ADD = 000000

W=0, 8bit
W=1, 16bit

MOD	Function
00	No displacement
01	Displacement (8bit)
10	" 16 bit
11	R/M is a reg

D=0, R/M, Reg
D=1, Reg, R/M
Destination Source

Code	Reg		R/M
	W=0	W=1	
000	AL	AX	[BX+SI]
001	CL	CX	BX+DI
010	DL	DX	BP+SI
011	BL	BX	BP+DI
100	AH	SP	SI
101	CH	BP	DI
110	DH	SI	BP
111	BH	DI	BX

Assembly to machine language

1. Register Addressing → MOV AX, BX

D W		MOD	Reg	R/M
100010	1 1	11	000	011

2. Direct Addressing → MOV, [1200H], BX

D W				
100010	0 1	00	011	110

4. Reg Indirect → MOV [BX], AL

100010	0 0	00	000	111

5. Base plus index → MOV CX, [BX+SI]

100010	1 1	00	001	000

6. Reg Relative → MOV CL, [BX+1000H]

100010	1 0	10	001	111	0000	0000	0001	0000

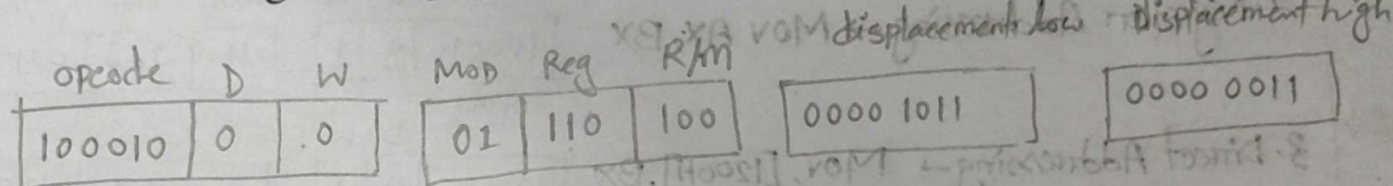
7. Base Relative plus Index → MOV [BP+DI+2500H], DX

100010	0 1	10	010	011			

Index

Example:

1. MOV [SI+3B], DH



Hexa: 8874B3

1. Reg Indirect $\rightarrow$ MOV [BX], AL

2. Base plus index $\rightarrow$ MOV CX, [BX+SI]

3. Reg Relative $\rightarrow$ MOV CL, [BX+1000H]

4. Base Relative plus index $\rightarrow$ MOV [BX+DI+2000H], DX

Machine to Assembly

Ex: 8A15H

opcode	D	W
100010	1	0

MOD	REG	R/M
00	010	101

Here, opcode = MOV

D = 1 ; Reg $\leftarrow$ R/M

W = 0 ; 8 bit Reg

MOD = 00 ; No displacement

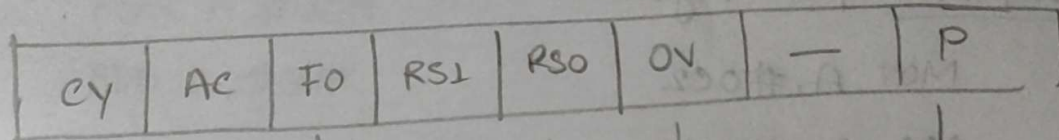
REG = 010 ; DL

R/M = 101 ; [DI]

we get, MOV DL, [DI]

Microcontroller

Chp-2



Reg Bank

8 bit 57 2(8) num of 1

odd P=1

even, P=0

low level carry High level carry A=1

Carry start CY=1

at start CY=0

$$\begin{array}{r}
 10111100 \leftarrow 98 \\
 11111111 \\
 \hline
 10000000 \leftarrow 101 \\
 00000000 \leftarrow 2
 \end{array}$$

1 = 99

1 = 94

0 = 07 629

0 = 129

0 = 029

0 = 10

0 = 9

Reg Bank

RS1	RS0	Register Bank
0	0	0
0	1	1
1	0	2
1	1	3

Example: PSW Register

MOV A, #0C2

ADD A, #3DH

ADD A, #01H

C2 → 1100 0010

3D → 0011 1101

1111 1111

+01 → 0000 0001

1 | 0000 0000

CY = 1

AC = 1

~~PSW~~ F0 = 0

RS1 = 0

RS0 = 0

OV = 0

P = 0

Ex: MOV A, #9CH
ADD A, #64H

$$\begin{array}{r}
 9C \rightarrow 1001 \overset{1}{1}100 \\
 64 \rightarrow 0110 \ 0100 \\
 \hline
 10000 \ 0000
 \end{array}$$

CY = 1

AC = 1

FO = 0

RS1 = 0

RS0 = 0

OV = 0

P = 0

Ex: MOV A, #0BFH
~~MOV~~ ADD A, #1BH

$$\begin{array}{r}
 BF \rightarrow 1011 \overset{1}{1}111 \\
 1B \rightarrow 0001 \ 1011 \\
 \hline
 1101 \ 1010
 \end{array}$$

CY = 0

AC = 1

FO = 0

RS1 = 0

RS0 = 0

OV = 0

P = 1

Physical Address

Physical Address

Given,
CS = 1111H
DS = 3333H
SS = 2526H
IP = 1232H
SP = 1100H
DI = 0020H

Calculate the corresponding physical address for the address bytes in CS, DS and SS.

Solve:

for CS, offset will be IP

$$PA = (\text{Segment} \times 10) + \text{offset}$$

$$= 1111 \times 10 + 1232$$

$$= 12342$$

for DS, offset will be DI

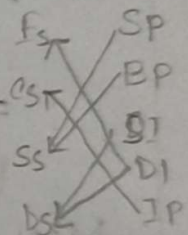
$$PA = 3333 \times 10 + 0020$$

$$= 33350$$

for SS, offset will be SP

$$PA = 2526 \times 10 + 1100$$

$$= 26360$$



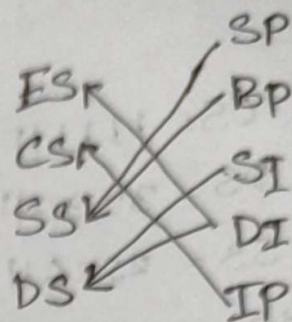
33.

DS = 1300H

SS = 1400H

BP = 1500H

SI = 0100H



MOV EAX, [BP+200H]

$$\begin{aligned} \text{PA} &= (\text{Segment} \times 10) + \text{offset} \\ &= (1400 \times 10) + (1500 + 200) \\ &= 15700\text{H} \end{aligned}$$

MOV AL, [BP+SI-200H]

$$\begin{aligned} \text{PA} &= (1400 \times 10) + (1500 + 0100 - 200) \\ &= 15400\text{H} \end{aligned}$$

MOV AL, [SI-0100H]

$$\begin{aligned} \text{PA} &= (1300 \times 10) + (0100 - 0100) \\ &= 13000\text{H} \end{aligned}$$

Q1. Starting address, ending address and physical address

A segment register contains DS = 3A00H and the offset address is 25F0H. Answer the ques.

i. Physical address:

$$\begin{aligned} & (\text{Segment} \times 10) + \text{offset} \\ &= (3A00 \times 10) + 25F0 \\ &= 3C5F0H \end{aligned}$$

ii. Starting address: $(\text{Segment} \times 10)$
 $= 3A00 \times 10 = 3A000H$

iii. Ending address: $(\text{Starting address} + FFFFH)$
 $= 3A000 + FFFF$
 $= 49FFF$

iv. If the physical address is 4C7A2H and the segment address is 4200H, calculate the offset address.

$$\begin{aligned} \text{Offset} &= \text{Physical Address} - (\text{Segment} \times 10) \\ &= 4C7A2 - 42000 \\ &= A7A2H \end{aligned}$$

v) If the offset address is 0A2BH and the physical address is 6B2BH, calculate the segment address.

$$\text{Segment address} = \frac{\text{Physical address} - \text{offset}}{10}$$

$$= \frac{6B2B - 0A2B}{10}$$

$$= \frac{6100}{10}$$

$$= 610H$$

$$= 610H$$

Stack

1. Show the stack and stack pointer for the following.

Assume the default stack area and register 0 is selected.

MOV R6, #25H
MOV R1, #12H
MOV R4, #0F3H
PUSH 6
PUSH 1
PUSH 4

POP 3
POP 1
POP 3

Solve!

Empty stack

0B	
0A	
09	
08	

SP=07

After:
PUSH 6

0B	
0A	
09	
08	25

SP=08

After PUSH 1

0B	
0A	0F3
09	12
08	25

SP=09

After PUSH 4

0B	
0A	0F3
09	12
08	25

SP=0A

After pop 3

0B	
0A	
09	12
08	25

SP=09

R3 = 0F3H

After POP 1

0B	
0A	
09	
08	25

SP=08

R1 = 12

After PUSH 3

0B	
0A	
09	0F3
08	25

SP=09

Example: Show the status of the CY, AC and P flag after the addition of 9CH and 64H in the following instructions.

MOV A, #9CH

Register Bank = 3

ADD A, #64H ;

9CH → 1001 1100

64H → 0110 0100

10000 0000 1011

CY = 1

AC = 1

FO = 0

RSW1 = 1

RSW0 = 1

OV = 1

PSW1 = 0

P = 0 even = 0

CY	AC	FO	RSW1	RSW0	OV	-	P
1	1	0	1	1	1	0	0

Show the contents of the PSW register after the execution of the following instructions.

MOV A, #0BFH

ADD A, #1BH

	D6	D5	E1	D4	
BF →	1011	1111	1001	→	0CH
1B →	0001	1011	0110	→	44H
	1101	1010	0000	1	

CY = 0

Ac = 1

F0 = 0

RS₀ = 0

RS₁ = 0

OV = 0

PSW1 = 0

P = 1 ; odd num of 1

Thus, the contents of the PSW is thus 01000001 ie 41H

MOV A, #0FFH

ADD A, #01

MOV A, #0C2H

ADD A, #3DH

PSW

PSW7	PSW6	PSW5	PSW4	PSW3	PSW2	PSW1	PSW0
CY	AC	FO	RS1	RS0	OV	-	P

CY = Carry flag

AC = Auxiliary carry flag

FO = Available to the user for general purpose

RS1 = Register Bank selection bit 1

RS0 = Register Bank selection bit 0

OV = overflow flag

-(PSW1) = User-definable bit

P = Parity flag

<u>RS1</u>	<u>RS0</u>	<u>Register Bank</u>	<u>Address</u>
0	0	0	00H - 07H
0	1	1	08H - 0FH
1	0	2	10H - 17H
1	1	3	18H - 1FH

b) Add the numbers 56H and 95H and show how the CY, AC and P flags are affected.

ADD the numbers 56H and 95H

$$\begin{array}{r} 56H \rightarrow 0101\ 0110 \\ 95H \rightarrow 1001\ 0101 \\ \hline 1110\ 1011 \end{array}$$

CY	AC	FO	RS1	RS0	OV	T	P
----	----	----	-----	-----	----	---	---

CY = 0

AC = 0

FO = 0 default

RS1 = 0

RS0 = 0

OV = 0 default overflow

PSW1 = 0 Default

P = 0 even = 0
odd = 1

ROM memory Address

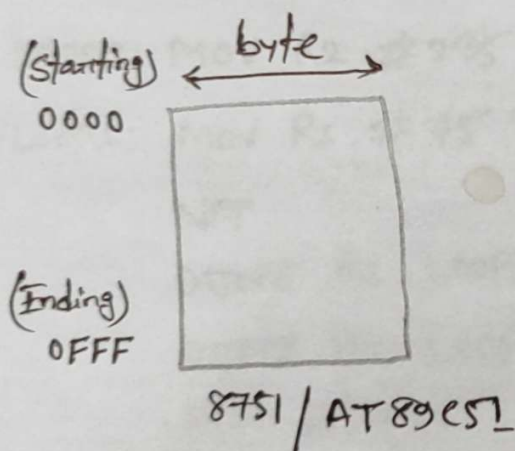
Find the ROM memory address of each of the following 8051 chip:

a) AT89C51 (or 8751) with 4KB

b) PS5000-32 with 32KB

a) ^{for} 4KB on chip ROM $4 \times 1024 = 4096 D$
 $= 1000H$

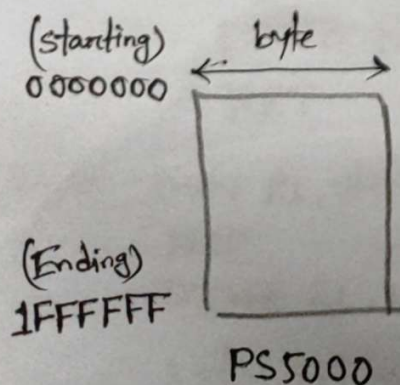
$= 1000H - 1$ (for limit)
 $= 0FFF$



b) ^{for} 32MB on chip ROM $32 \times 1024 \times 1024 = 33554432 D$

$= 2000000H - 1$

$= 1FFFFFFFH$



Delay Math

Exercise: Find the period of the machine cycle for 11.0592 MHz cycle frequency.

$$F = 11.0592 \text{ MHz}$$

$$\text{internal frequency, } f = \frac{11.0592}{12} = 0.9216 \text{ MHz}$$

$$T = \frac{1}{f} = \frac{1}{0.9216} \mu s = 1.085 \mu s$$

```

Loop3: Mov R3, #150
Loop2: Mov R2, #235
Loop1: Mov R1, #75
      NOP
      DJNZ R1, LOOP1
      DJNZ R2, LOOP2
      MOV A, #25
      MOV B, #10
      MUL AB
      DJNZ R3, LOOP3
      RET
  
```

Ans:

Machine cycle:

$$8038353 \times 1.085 = 8721613.005 \mu s$$

loop1: MOV R1, #75 — 1
 NOP — 1
 DJNZ R1, LOOP1 — 2
 $1 + (3 \times 75) = 226$

loop2: MOV R2, #75 → 1
 LJ — 226
 DJNZ R2, LOOP2 → 2
 $1 + (228 \times 235) = 53581$

loop3: MOV R3, #150 — 1
 LJ — 53581
 MOV A, #25 — 1
 MOV B, #10 — 1
 MUL AB — 4
 DJNZ R3, LOOP3 — 2
 $3 + (53581 \times 150) = 8038353$

Example:

MOV A, #55H

MOV R1, A

ACALL DELAY

CPL A

SJMP AGAIN ;

DELAY: MOV R3, #200

HERE: DJNZ R3, HERE

RET

Solve:

MOV R3, #200 — 1

DJNZ R3, HERE — 2 } 2x200

RET — 2

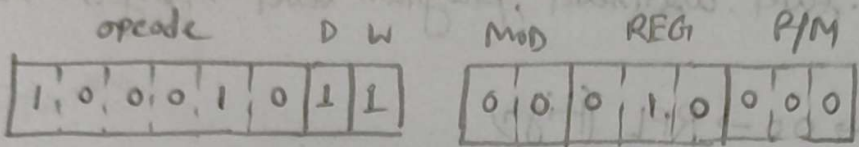
$$\frac{12}{11.0592} = 1.085 \mu s$$

$$\therefore \text{Time delay} = 403 \times 1.085 = 437.255 \mu s$$

Assembly $\Rightarrow$ Machine

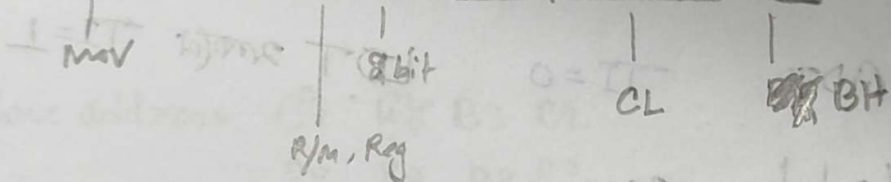
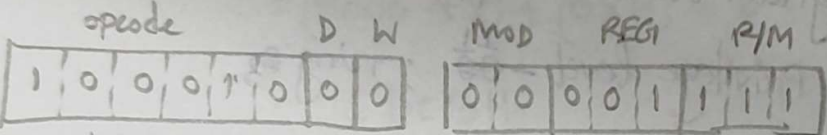
Example:

MOV DX, [BX+SI]



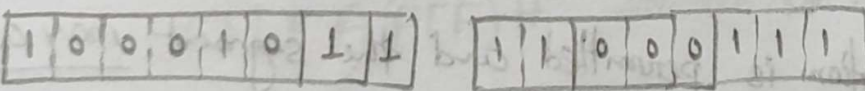
0 0 0	AX	CX+SI
0 0 1	CX	BX+DI
0 1 0	DX	BP+SI
0 1 1	BX	BP+DI
1 0 0	SP	SI
1 0 1	BP	DI
1 1 0	SI	BP
1 1 1	DI	BX

880F

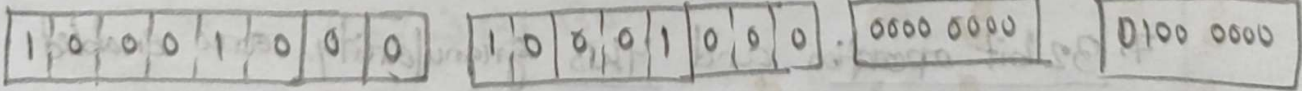


MOV [BH], CL

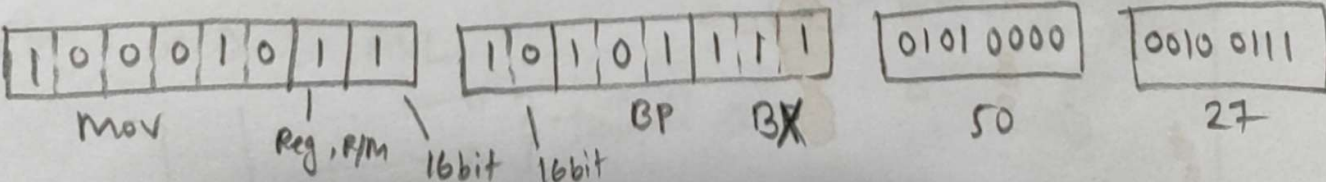
MOV AX, [DI]



MOV [BX+SI+4000H], CL



8BAF5027H



MOV BP, [BX+2750H]

Protected mode

Kernel, operating system, OS, supervisor level $\rightarrow$ Ring 0 $\rightarrow$ 00

Application level, user level, program level $\rightarrow$ Ring 3 $\rightarrow$ 11

Ring $\rightarrow$ 00 $\rightarrow$ RPL (High)

1. The program works in application level

Privilege level $\rightarrow$ ~~00~~¹¹ ; Ring 3

2. Use GDT $\quad$ $\text{TI} = 0$ $\quad$ LDT $\text{TI} = 1$

3. Index/selector 3000

4. Base address $\begin{array}{c} \text{B7} \text{ B4} \text{ B3} \text{ B2} \\ \hline 00 \text{ } 80 \text{ } 00 \text{ } 00 \text{ H} \end{array}$

5. The segment is valid code segment $S = 1$

6. Code execution is permitted and the segment may also be read
code execution $\rightarrow E = 1$ $R/W = 1$

7. 32 bit operation and 4KB granularity are enabled.
 $D = 1$ $G = 1$

8. The privilege level of the code segment consistent with the privilege level of the executing application.

Solve:

$$\begin{aligned}
 1) \text{ Segment value} &= (\text{Index} \times 8) + (\text{TI} \times 4) + \text{RPL} \\
 &= (3000 \times 8) + (0 \times 4) + 3 \\
 &= 5DC3H
 \end{aligned}$$

$$\begin{aligned}
 \text{Base address} &= B7 \ B4 \ B3 \ B2 \\
 &= 00 \ 80 \ 00 \ 00
 \end{aligned}$$

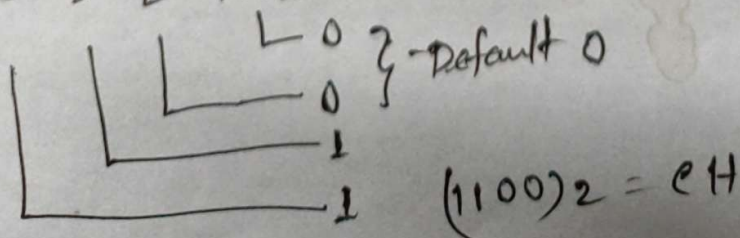
We know ending FFFFFFFF

$$\begin{aligned}
 \therefore \text{limit} &= \text{FFFFFFFF} - 0080000 \\
 &= \text{FF7FFFFF}H \rightarrow \text{এই সারদ দিবে - যদি 8 bit গ্রুপে} \\
 &= \text{FF7FFF limit}
 \end{aligned}$$

$$\begin{aligned}
 \text{Limit address} &= \underline{B6} \ \underline{(L)} \ \underline{B1} \ \underline{B0} \\
 &= \underline{F} \ \underline{F7} \ \underline{FF}
 \end{aligned}$$

Flag = B6 (Higher)

Flag = G D L AVL



1. Suppose that $DS = 0250H$, $BX = 0370H$ and $DI = 400H$.

Determine the memory address accessed by each of the following instructions, assuming real mode operation.

a) $MOV AL, [3154H]$

b) $MOV [DI], AL$

c) $MOV DL, [BX + DI]$

a) this is a Direct Addressing mode

$MOV AL, [3154H]$

Given,

$DS = 0250H$

$BX = 0370H$

$DI = 400H$

We get, $DS = 0250H$

offset = $3154H$

$$\begin{aligned}\therefore \text{Physical address} &= (\text{Segment} \times 10H) + \text{offset} \\ &= (0250 \times 10) + 3154 \\ &= 5654H\end{aligned}$$

b) $MOV [DI], AL$ is Register Indirect Addressing Mode.

$$\begin{aligned}\therefore P.A &= (\text{Segment} \times 10H) + \text{offset} \\ &= (0250 \times 10) + 400 \\ &= 2900H\end{aligned}$$

c) MOV DL, [BX+DI]

$$\begin{aligned} \text{P.A} &= (\text{Segment} \times 10) + \text{offset} \\ &= (0250 \times 10) + (0370 + 400) \\ &= 2C70 \text{ H} \end{aligned}$$

2. In the real mode, show the starting and ending addresses of each segment located by the following segment register values

a) 1000H

b) 1234H

a) 1000H

$$\begin{aligned} \text{Starting address} &= \text{Segment} \times 10\text{H} \\ &= 1000 \times 10\text{H} \\ &= 10000\text{H} \end{aligned}$$

$$\begin{aligned} \text{Ending address} &= \text{Starting} + \text{FFFFH} \\ &= 10000 + \text{FFFFH} \\ &= 1FFFF\text{H} \end{aligned}$$

b) 1234H

$$\begin{aligned} \text{Starting address} &= \text{Segment} \times 10 \\ &= 1234 \times 10 \\ &= 12340 \end{aligned}$$

$$\begin{aligned} \text{Ending address} &= \text{Starting} + \text{FFFFH} \\ &= 12340 + \text{FFFFH} \\ &= 2233\text{F} \end{aligned}$$

3. Convert the following:

a) Assembly to machine: `MOV BX, SS:[BP+DI]`

b) machine to assembly: `8B07H`

000
001
010
011

a) `MOV BX, SS:[BP+DI]`

opcode	D	W	MOD	REG	R/M
1000010	1	1	00	01	011

we get:

opcode = `MOV`

$D=1$; $Reg \leftarrow R/M$

$W=1$; 16 bit

$MOD=00$; No displacement

$REG=BX=011$

$R/M=011$; `[BP+DI]`

In Hexa we get `8B1B`

b) `8B07H`

opcode	D	W	MOD	REG	R/M
1000010	1	1	00	00	111

We get, opcode = `MOV`

$D=1$; $Reg \leftarrow R/M$

$W=1$; 16 bit

$MOD=00$; No displacement

$REG=000$; `AX`

$R/M=111$; `BX`

$\therefore$ `MOV AX, [BX]`

Chp-3

Delay Calculation

- Find the period of the machine cycle for 11.0592 MHz cycle frequency.

Loop3: MOV R3, #150 150 x 1T
 Loop2: MOV R2, #235 235 x 1T
 Loop1: MOV R1, #75 75 x 1T
 NOP
 DJNZ R1, Loop1 235 x 150
 DJNZ R2, Loop2 150 x 235
 MOV A, #25
 MOV B, #10
 MUL A, B 150 x 235
 DJNZ R3, Loop3
 RET

loop1: MOV R1, #75 — 1T x 1
 NOP — 1T
 DJNZ R1, loop1 — 2T
 1 + (3 x 75) = 226
 226 x 235 x 150
 = 7966500

loop2: MOV R2, #235 — 1
 loop1 — 226
 DJNZ R2, loop2 — 2
 226 + 2 = 228 x 235
 1 + (228 x 235)
 = 53580 + 1 = 53581

Loop3: MOV R3, #150 — 1T

Loop2 — 53581

Loop1

MOV A, #25 — 1T

MOV B, #10 — 1T

MUL A, B — 3T

DJNZ R3, Loop3 — 2T

RET — 2T

$$3 + (53588 \times 150)$$

$$= 8038203$$

Given, $F = 11.0592 \text{ MHz}$

$$\text{internal frequency, } f = \frac{11.0592}{12} = 0.9216$$

$$T = \frac{1}{f} = \frac{1}{0.9216} \mu s = 1.085 \mu s$$

$$\text{Delay} = 8038203 \times 1.085 = 8.721613$$

$$= 8721613.005 \mu s$$

Example: (3.16)

DELAY: MOV R2, #250 — 1

HERE: NOP

NOP

NOP

NOP

DJNZ R3, HERE

RET — 2

$$(3 + 6 \times 250) \times 1.085$$

$$=$$

Time Period

* MOV — 1

NOP — 1

DJNZ — 2

Example:

MOV R2, #19 — 1

L1: MOV R3, #255 — 1

L2: MOV R4, #55 — 1

L3: DJNZ R4, L3 — 2

DJNZ R3, L2 — 2

DJNZ R2, L1 — 2

RET — 2

for L3: MOV R2, #19

L2

DJNZ R2, L1

RET

$$3 + (28816 \times 19)$$

$$= 547507^{45} \times 1.085$$

= .

$$L3: 1 + (2 \times 255) = 511$$

MOV R4, #55

DJNZ R4, L3

L2: MOV R3, #255

Loop 3

DJNZ R3, L2

$$1 + (513 \times 255) = 28816$$

Microprocessor

8086

8086 is a 16 bit microprocessor, because it can process 16 bit data at a time.

Ex: AX, BX, CX, DX [16 bit registers]

16 bit value means $2^{16} = 65536$

So, it can represent values from 0 to 65535

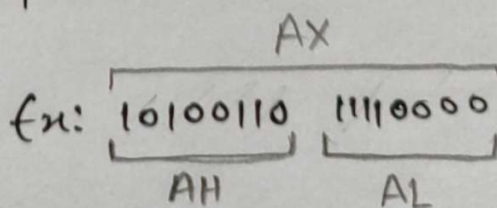
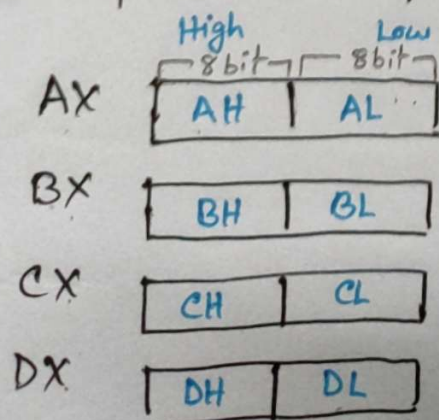
Types of Registers

8086 has 14 registers. Per register 16 bit

These registers can be divided into 3 main groups.

1. Data Registers
2. Address Registers
3. Status / Flag registers

1. Data registers: It is called general purpose register because they are used for data operation.



2. Address Register: it is used to hold memory address

3 types:

1. Segment registers (CS, DS, ES, SS)
 code Data Extra stack
2. Pointer registers (SP, BP, IP)
 stack Base Instruction
3. Index registers (SI, DI)
 Source Destination

CPU needs Address, Data to access memory.

Size of memory address bit = block size

Maximum memory address $2^{20} = 1048576$ bytes

$\approx 10^6$

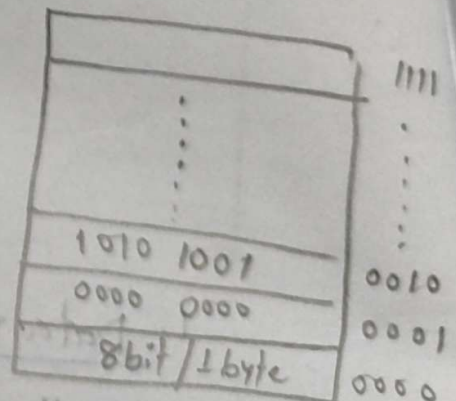
= 1 MegaByte

$\therefore$ 8086 can access 1 MB memory

20 bit binary value in Hexadecimal

$\frac{1111}{F}$, $\frac{1010}{A}$, $\frac{1100}{C}$, $\frac{0000}{0}$, $\frac{1011}{B}$

= FAC0B(H) $\rightarrow$ Hexa



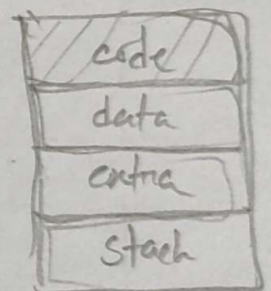
1MB

AX

BX

8086 has 4 main segment registers:

<u>Register</u>	<u>Functions</u>
CS (Code segment)	Holds program/code
DS (Data segment)	Holds data
ES (Extra segment)	Extra data area
SS (Stack segment)	Holds stack memory



C-0
22 June 20

chp-3 Delay Calculation

→ Code execute 20

Function (nested loop)

DELAY: MOV R1, #50 → 1 cycle

LABEL1: MOV R2, #100 → 50 inner loop

LABEL2: NOP (no operation) → 50x100 = 5000 outer loop 50 times

DJNZ R2, LABEL2 → 5000 outer loop counter x inner loop counter

DJNZ R1, LABEL1 → 50

RET → 1

Loop

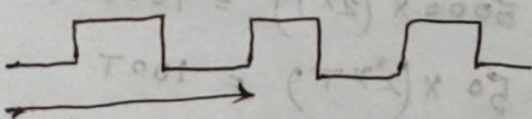
MOV R2, #5

L:

DJNZ R2, L }

Machine cycle -

CPU executing an instruction takes a certain number of clock cycles.
→ These are referred as to as machine cycles.



T = time period

$$f = \frac{1}{T} \quad T = \frac{1}{f}$$

clock wave, continuous

Crystal oscillator (20 or frequency)
machine (1)

Oscillator is $\frac{1}{12}$ frequency machine
8051

12 MHz → 1 MHz (8051)

Time period

MOV $\rightarrow 1$

MUL, DIV $\rightarrow 4$

NOP $\rightarrow 1$

DJNZ $\rightarrow 2$

RET $\rightarrow 2$

*

* 121

Find the period of the machine cycle for 11.0592 MHz cycle frequency.

$$F = 11.0592 \text{ MHz}$$

$$\begin{aligned} \text{internal frequency, } f &= \frac{11.0592}{12} \\ &= 0.9216 \text{ MHz} \end{aligned}$$

$$T = \frac{1}{f} = \frac{1}{0.9216} \mu s = 1.085 \mu s$$

THE

DELAY: MOV R1, #50

$$1 \times (1 \times T) = 1T$$

LABEL1: MOV R2, #100

$$50 \times (1 \times T) = 50T$$

LABEL2: NOP

$$50 \times 1000 = 5000 \times (1 \times T) = 5000T$$

DJNZ R2, LABEL2

$$5000 \times (2 \times T) = 10000T$$

DJNZ R1, LABEL1

$$50 \times (2 \times T) = 100T$$

RET

$$1 \times (2 \times T) = 2T$$

$$= 15153 \times T$$

$$= 15153 \times 1.085$$

$$= 16441.005 \mu s$$

DELAY: MOV R0, #200 — ~~200~~ × (1×T) = ~~200~~ T 1T

HERE: DJNZ R0, HERE — 200 × (2×T) = 400 T

RET — 1 × (2×T) = 2 T

$$= 603 \times T$$

$$= 653.17 \mu s$$

$$\left[(200 \times 2) + 1 + 2 \right] \times 1.085 = 437$$

— x — 280.1 × 100000

DELAY: MOV R3, #250 — 1 T

HERE: NOP — 250 × (1×T)

NOP — 250 × (1×T)

NOP — 250 × (1×T)

NOP — 250 × (1×T)

DJNZ R3, HERE — 250 × (2×T)

RET — 2 T

=

$$1503 \times 1.085$$

$$= 1630.755$$

08
18 June 26

Access Right 80286

P	DPL	S	E	ED/c	R/W	A
1	11	1	0	1	0	0

a) DPL = 11 (lower) segment access will not be granted

b) P = 1

c) S = 1

d) E = 0

e) ED = 1 expanded downward

f) W = 0

A = 0

Ans: 111101001

(x:p.88, 89)

8051 Stack :

memory এর একটি অংশ stack এর মাধ্যমে কাজ করে

Stack pointer (SP)

↳ currently stack এর top 1 কে অর্থাৎ, top position

SP is default value, $SP = 07H$

↓
value change করতে পারবে, default value use করতে পারবে।

stack pointer value = $07H$ (Default)

Stack এর শুরু $07H + 1 = 08H$

Empty stack এর মতো

0BH	
0AH	
09H	
08H	

starting -

$SP = 07H$

PUSH & POP operation (R0 .. R7) রেজিস্টার গুলো use হবে।

Syntax: PUSH 1 → R reg এর variation হওয়া

(PUSH RL)

1 = R1

Push operation

Empty stack

07H	
0AH	
09H	
08H	

SP = 07H

After push 6

0BH	
0AH	
09H	
08H	25H

SP = 08H

MOV R6, #25H

MOV R4, #12H

MOV R4, #0F3H

PUSH 6

PUSH 2

PUSH 4

POP 3

POP 1

PUSH 3

After push 1

0BH	
0AH	
09H	12H
08H	25H

SP = 09H

After push 4

0BH	
0AH	0F3
09H	12
08H	25H

SP = 0AH

* Top value = SP

Pop operation

sta Syntax: POP 3 → Top element delete & R3 prev value

H80 R3/Reg 1 store karke

(processor specific)

After POP 3

0AH	
09H	12H
08H	25H

SP = 09

R3 = F3H

* push → first SP increment karke value rahi

* POP → first value remove krke then SP value decrement krke

After POP 1

0AH	
09H	
08H	25H

SP = 08

R1 = 12H

After PUSH 3

0AH	
09H	F3H
08H	25H

SP = 09H

1

Page 103 Example

MOV SP, #5FH

MOV R2, #25H

MOV R1, #12H

MOV R4, #0FBH

PUSH 2

PUSH 1

PUSH 4

Initial state

64H	
62H	
62H	
60H	

SP = 5FH

After PUSH 2

62H	
62H	
60H	25H

SP = 60H

After PUSH 1

62	
62	12H
60	25H

SP = 62

After PUSH 4

62	F3
62	12H
60	25H

SP = 62

8051 : Time delay calculation

Loop I

loop action is performed by

DJNZ reg, label

Decrement
reg

Jump if Not Zero
≠ 0

MOV R1, #10

Label: {

DJNZ R1, Label

MOV R1, #10

→ 1 st execute 250

Label: {

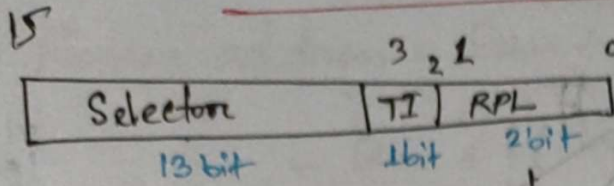
MOV A, #55H

MOV R3, A

DJNZ R1, label

→ 10 st execute 250

Protected mode (memory addressing) ~~Math~~



Request Privileged Level

00 = High

11 = Low

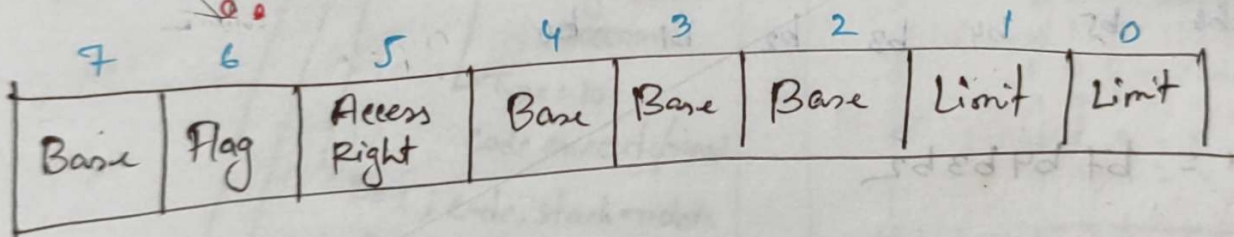
Get Description Table

0 = Global

1 = Local

Selector

$2^{13} = 8192$ either in Global/Local



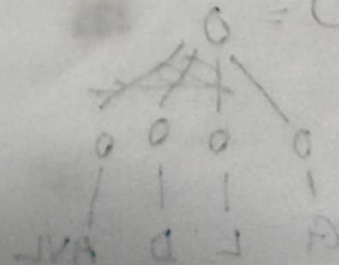
Base / Starting

Limit

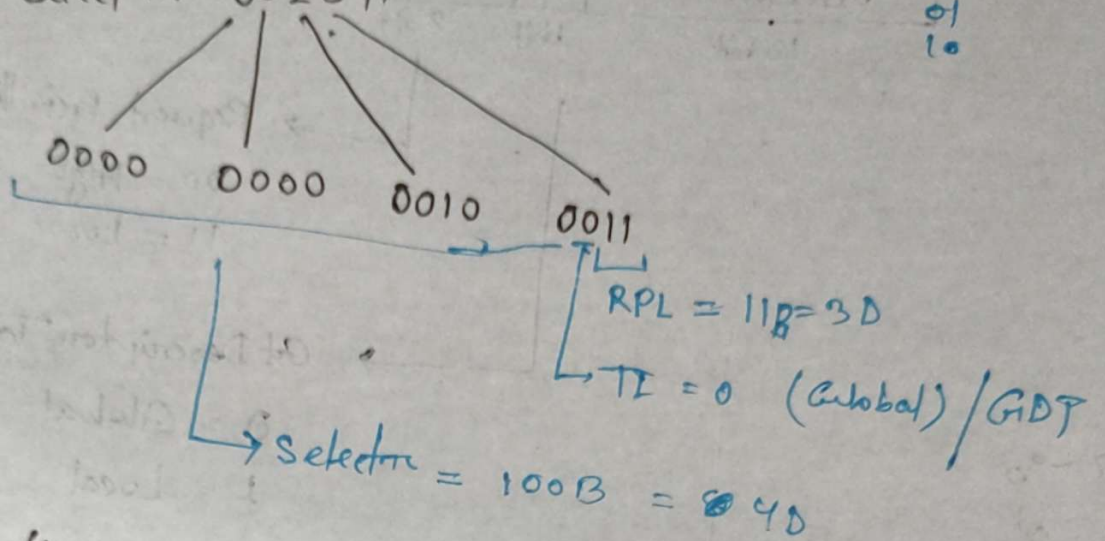
Ending address

Access right

Flag = 0 (High)



i) Segment Selector: 0023H



ii) Selector = 4, so, Descriptor = 4

00 00 FA 34 00 20 1A BC H
b7 b6 b5 b4 b3 b2 b1 b0

Base = b7 b4 b3 b2

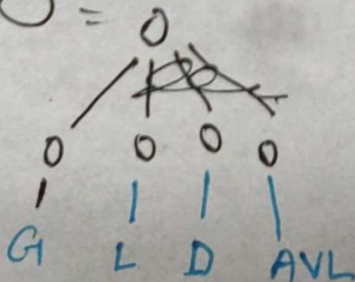
= 00340020H

Limit = b6 (lower) b1 b0

= 01ABC H

Ending Address:

Flag = B6 (Higher)



G = Segment Size (32/64/16/8)

L = Instruction Size (16/32)

D = Default

AVL = 0 = Segment available

1 = Not "

P | DPL | S | Type | A

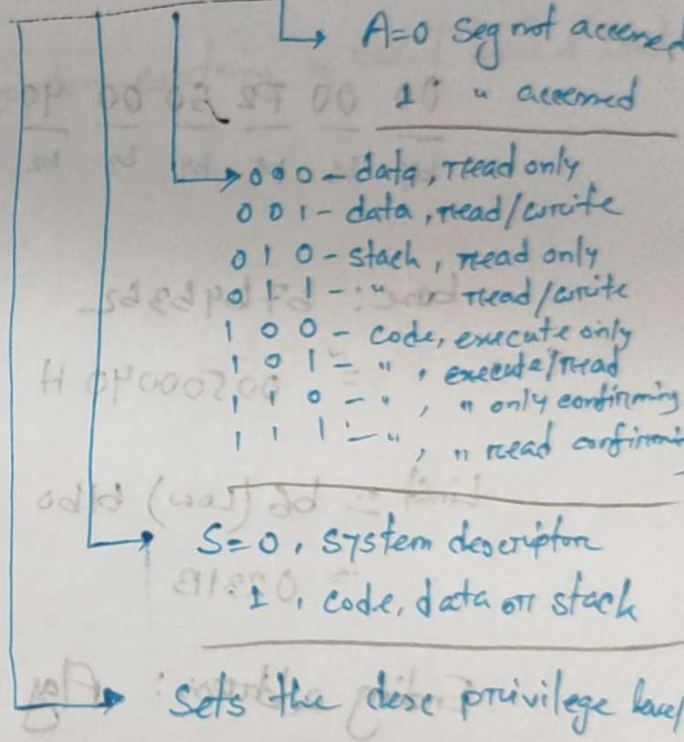
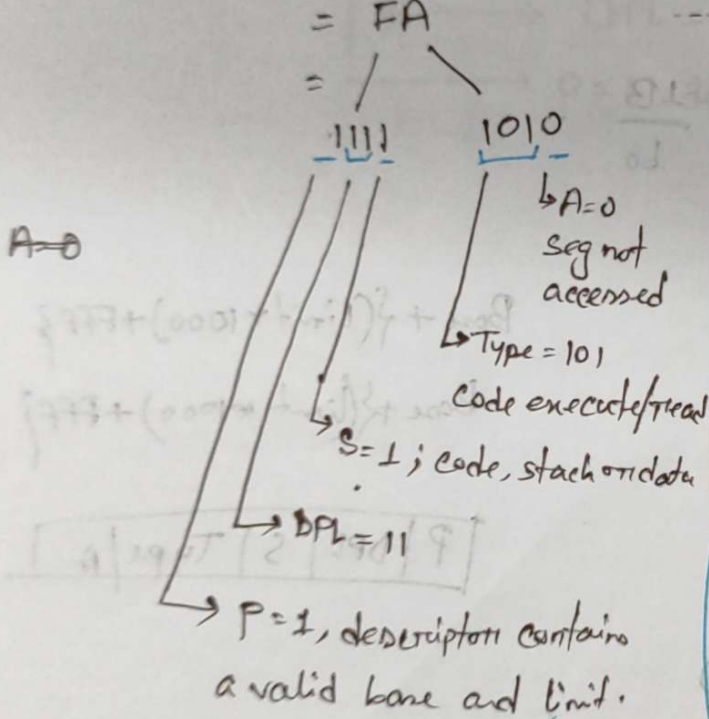
$G=0$, Ending address = Base + Limit

$G=1$, " = Base + $\{ (Limit \times 1000H) + FFFFH \}$

Base + Limit = 00340020 + 01ABE

iv) Access Right = BS

P	DPL	S	Type	A
1bit	2bit	1bit	3bit	1bit



v) Segment type; 101,
Code execute/read

P=0, descriptor is undefined

1, " contains a valid base and limit.

Ques 4

Segment Selector = 0031H

0000 0000 0011 0001

LRPL = 01 = 1

TI = 0 (GDT)

Selector = 7

i) Selector = 7 so, Description 7

00 00 F2 50 00 40 23 1B
b7 b6 b5 b4 b3 b2 b1 b0

base: b7 b6 b5 b4

= 00500040 H

Limit = b6 (Low) b1 b0

= 0231B

Base + {(Limit * 1000) + FFF}

Base + {(Limit * 1000) + FFF}

[P | DPL | S | Type | A]

Ending address: Flag = 0

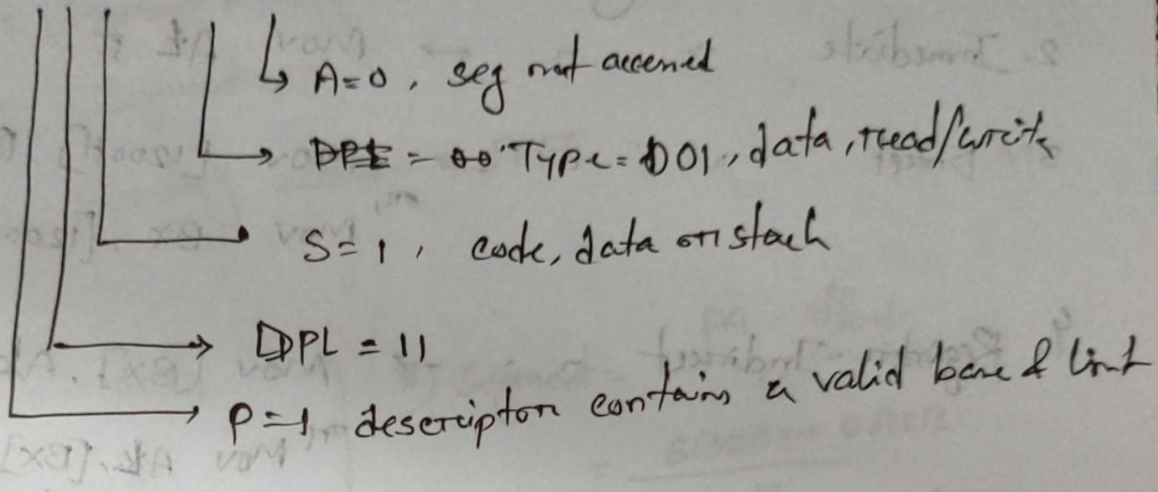
0000
G L D AVAL

G=0, B+L = 00500040 + 0231B

P	DPL	S	Type	A
---	-----	---	------	---

iii) Access Right = b5
= F2

1111 0010



$$1. \text{ DS} = 3A00 \text{ H}$$

$$\text{offset} = 25F0 \text{ H}$$

$$\boxed{\begin{aligned} \text{Physical Address} &= (\text{Segment} \times 10 \text{ H}) + \text{offset} \\ \text{Starting} &= \text{Segment} \times 10 \text{ H} \\ \text{Ending Address} &= \text{Starting address} + \text{FFFF} \end{aligned}}$$

$$i) \text{ PA} = 3A00 \times 10 + 25F0$$

$$= 3C5F0 \text{ H}$$

$$ii) \text{ Starting} = \text{Segment} \times 10 \text{ H}$$

$$= 3A00 \times 10$$

$$= 3A000 \text{ H}$$

$$iii) \text{ Ending address} = \text{Starting} + \text{FFFF}$$

$$= 3A000 + \text{FFFF}$$

$$= 40FFF \text{ H}$$

$$iv) \text{ offset} = \text{PA} - (\text{Segment} \times 10)$$

$$= 4C7A2 - 42000$$

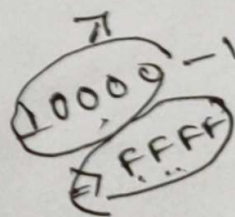
$$= A7A2$$

$$v) \text{ Segment} = \frac{\text{PA} - \text{offset}}{10}$$

$$= \frac{6B2B3 - 0A2B}{10}$$

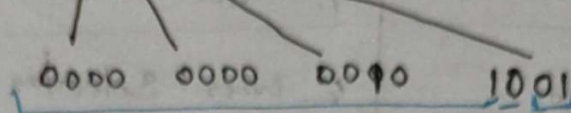
$$= \cancel{0688} \text{ H } 6A88 \text{ H}$$

$$\begin{aligned} &(\text{PA} - 1) \\ &= (3C5F0 \text{ H} - 1) \\ &= \text{End} \end{aligned}$$



Protected mode

Segment Selector = 00 20



Selector = 5

i) Descriptor Table = \$ GDT

RPL = 01 = 1

7	6	5	4	3	2	1	0
base	Flag	AR	B	B	B	L	L
00	00	F2	50	00	20	23	1B

ii) Descriptor index num = 5

iii) Base address = b7-b4b3b2

= 0050 0020 H

iv) Segment Limit = b6 (lower) b1 b0

= 0231B H

v) Ending address:

Flag = 0

= 0000

G L D AVL

G=0, EA = Base + Limit

= 0500020 + 0231B

= 50233B H

Real Mode

1. Access only 1MB memory
2. uses segment: offset addressing
3. X no memory Protection
4. no multitasking

Protected Mode

1. Access more than 1MB
2. uses Selector + Descriptors
3. provides memory Protection
4. Supports multitasking

Register Addressing : `MOV AX, BX`

`MOV AX, 1234H` → Immediate Addressing

`MOV AL, [123H]` → Direct Addressing

`MOV AX, [BX]` → Register Indirect

`MOV AX, [BX + SI + 100H]` → Base Relative Plus Index addressing

GDT

Global Descriptor Table
Contains segment definitions for all programs
System wide table

LDT

Local Descriptor Table
Contains segment definition for one application
Application specific table